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Conditional Cash Transfers as Stabilization Tool: a HANK Approach

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1. Introduction and Motivation

Fiscal transfers are one of the most widely used tools of redistributive policy. Brazil's Bolsa Família (BF) is a leading example: a Conditional Cash Transfer (CCT) targeted at low-income households, covering nearly a third of the population. During COVID-19, its emergency expansion (Auxílio Emergencial) is widely credited with cushioning the aggregate fall in output.

Yet, standard macroeconomic models cannot speak to this question. In a representative-agent framework, a lump-sum transfer is fully saved by Ricardian households and has no real effect (Galí 2015). The key insight of Heterogeneous Agent New-Keynesian (HANK) models is that households differ in their Marginal Propensity to Consume (MPC), which means that transfers concentrated on liquidity-constrained households generate substantially larger aggregate demand effects than uniform transfers of equal budget cost.

This paper develops a HANK model calibrated to Brazil to study CCTs as a stabilization tool. The economy features three labor market states: formal and informal employment, and unemployment, with idiosyncratic productivity risk and borrowing constraints. Only informal workers below an income threshold and the unemployed receive BF, matching the program's actual targeting rule.

We solve the model using the Sequence-Space Jacobian (SSJ) and compute both partial- and general-equilibrium responses of aggregate consumption to a shock in the transfer level T. The main object of interest is dC/dT: the intertemporal consumption response to the CCT.

We find that BF recipients are disproportionately hand-to-mouth, yielding high impact MPCs and fast expenditure decay. GE amplification through wages and interest rates is currently modest, a result we discuss later.

Literature Review

Our framework builds on the incomplete-markets tradition initiated by Aiyagari (1994), in which uninsurable idiosyncratic risk and borrowing constraints generate a non-degenerate distribution of wealth. In the representative-agent New-Keynesian model, all households are alike and Ricardian equivalence holds: a lump-sum transfer financed by future taxes is fully saved and has no real effect (Galí 2015).

However, microdata show that households hold very different MPCs depending on their income and credit constraints (Kaplan and Violante 2014). Kaplan, Moll, and Violante (2018) embed this heterogeneity in a New-Keynesian environment, coining the term HANK and showing that a sizeable share of wealthy hand-to-mouth (HtM) households rationalizes the high MPCs observed in the data.

In these models, Ricardian equivalence breaks down for three reasons: binding borrowing constraints, precautionary saving against uninsurable risk, and the fact that the households who receive a transfer need not be the ones who pay for it. Therefore, fiscal policy has real and distributional effects, and the size of those effects depends on who receives the transfer (Auclert, Rognlie, and Straub 2025).

Methodologically, we rely on the Sequence-Space Jacobian (SSJ) of Auclert, Bardóczy, Rognlie, and Straub (2021), which summarizes the model as a system of equations in aggregate sequences and recovers general-equilibrium responses from a small set of Jacobians. This approach is well suited to our question: the object of interest is the consumption response to the conditional transfer, $dC = \mathbf{G}_{C,T}\, dT$.

The interpretation of this object is grounded in Auclert, Rognlie, and Straub (2024), who generalize the Keynesian cross to a dynamic setting and show that the matrix of intertemporal MPCs (iMPCs), together with the path of deficits, is a sufficient statistic for the fiscal multiplier.

Additionally, Auclert, Bardóczy, and Rognlie (2023) argue that only a New-Keynesian model with sticky wages can match the empirical MPCs, marginal propensities to earn (MPE), and fiscal multipliers simultaneously, which motivates our wage-rigidity block following Erceg, Henderson, and Levin (2000) and Hagedorn, Manovskii, and Mitman (2019).

A growing literature studies fiscal policy and aggregate demand with heterogeneous agents. Auclert and Rognlie (2018) were among the first to connect the distribution of income to aggregate demand, showing that inequality matters for output. Because targeted transfers concentrate resources on high-MPC households, they deliver a larger multiplier than a universal lump-sum transfer of the same budget.

Auclert, Rognlie, and Straub (2023) show that debt-financed transfers are spent quickly by high-MPC households and then trickle up to wealthier savers, generating a larger, more persistent, and more inflationary response than tax-financed transfers, while widening inequality. In the same spirit, Ferriere and Navarro (2025) find that the distribution of taxes is a key determinant of the spending multiplier and of its welfare consequences.

Our exercise inherits this logic and applies it to two financing regimes, one based on debt and another on taxes. Closer to our question, McKay and Reis (2016) study the role of automatic stabilizers in a HANK economy, and Bartal and Becard (2024) compare consumption tax cuts with lump-sum stimulus payments, the closest benchmark to our comparison of fiscal instruments. Krueger, Mitman, and Perri (2016) likewise study the distribution of welfare losses over large recessions.

Our policy instrument connects this macroeconomic literature to the study of Conditional Cash Transfers. The micro evidence is by now extensive: Fiszbein and Schady (2009) document that CCT programs are generally well targeted, raise consumption, and reduce poverty, with only modest effects on labor supply. This literature, however, is mostly microeconomic and focused on long-run human-capital outcomes.

A smaller but growing body of work studies the aggregate effects of cash transfers. Egger et al. (2022) exploit a large experimental transfer in Kenya, a fiscal shock of roughly 15% of local GDP, and estimate a local transfer multiplier of about 2.5, with sizeable spillovers to non-recipients and minimal price inflation.

For Brazil, Corbi, Papaioannou, and Surico (2019) estimate sizeable local multipliers from federal transfers. The COVID-19 episode is a natural motivation: the emergency expansion of transfers (Auxílio Emergencial) reached about a third of the population and is widely credited with cushioning the fall in output and curbing the rise in poverty and inequality during the crisis.

Taken together, the existing evidence on cash transfers is either microeconomic, reduced-form, or based on accounting and computable-general-equilibrium frameworks, none of which can compare targeting rules under a common budget, decompose aggregate from distributional effects through iMPCs, and discipline the short-run stabilization role of the policy with nominal rigidities.

To the best of our knowledge, this is the first structural HANK model to study a CCT (Bolsa Família) as a stabilization tool in a developing economy with a dual, formal-informal labor market.

2. Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady-state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity, labor sector, and discount factor.

Households enjoy a GHH utility over consumption $c_{i,t}$ and labor $h_{i,t}$ (Greenwood, Hercowitz, and Huffman 1988), defined as a CRRA utility $u(\tilde{c}_{i,t}) = u(c_{i,t} - v(h_{i,t}))$,

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta_{i,t}^t \frac{\tilde{c}_{i,t}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} \right], \quad \text{with } v(h) = \psi \frac{h^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}},$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

The GHH specification assumes a zero income effect on hours worked, so that the hours decision is separable from the consumption-savings problem, which seems to be a reasonable approximation for informal labor markets in development countries.

A household faces labor section risk $s \in \{F, I, U\}$, idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, where $\delta_\beta := \beta_H - \beta_L > 0$ and there is ω_I impatient households. The labor supply risk classifies households as belonging to the formal or informal sector, or as unemployed.

Formal workers earn labor income $(1-\tau_\ell) w_t e_{i,t} h_{i,t}$, where w_t is the wage rate, τ_ℓ is a fixed labor income tax rate. While, informal workers receive benefits $w_t^I e_{i,t} h_{i,t}$, where $w_t^I = \xi w_t$ and $\xi \in (0,1)$ is the wage gap. All unemployed and eligible informal households earn the benefit T_t. Let $y_t(s,e)$ be this non-financial income,

$$y_t(s,e) = \begin{cases} (1-\tau_t) w_t e, & \text{if formal;} \\ w_t^I e h_t + T_t \cdot \mathbb{1}[w^I e h_t < \bar{y}], & \text{if informal;} \\ T_t, & \text{if unemployed.} \end{cases}$$

We assume that formal wages are observable by the government, so that only informal workers and unemployed may be eligible to receive Bolsa Família (BF). We assume that informal households are eligible if $w_t^I e_{i,t} h_{i,t} < \bar{y}$ and calibrate this to match to the number of beneficiaries.

Households save into liquid assets $a_{i,t}$ with ex-ante real return $1+r_t = (1+i_{t-1})/(1+\pi_t)$, where i_t represents the nominal interest rate and π_t denotes inflation. They also receive dividends $d_{i,t}$ from firms, which we assume to be proportional to productivity $d_{i,t} = d_t \cdot e_{i,t}$[1]. Their budget constraint at time t reads:

$$c_{i,t} + a_{i,t} = (1+r_t) a_{i,t-1} + d_{i,t} + y_{i,t}(s,e).$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply formal households delegate their labor supply decisions to labor unions, and

[1]Dividends are proportional to productivity in order to increase inequality. We aim to perform robustness tests with other specifications.

thus take worked hours as given. While, informal ones choose hours to maximize their utility, thus, for each i,

$$\psi \left(h^I\right)^{1/\varphi} = w^I e \quad \Leftrightarrow \quad h^I(e) = \left[\frac{w^I e}{\psi}\right]^{\varphi}.$$

This generates a productivity cutoff for informal workers of

$$w^I e \left(\frac{w^I e}{\psi}\right)^{\varphi} < \bar{y} \quad \Leftrightarrow \quad e < e^* = \frac{(\psi^{\varphi} \bar{y})^{\frac{1}{1+\varphi}}}{w^I}.$$

We solved the household problem using the Endogenous Grid Method (Carroll 2006). The Bellman Equation for each individual i is:

$$\begin{aligned} V_t(s, \beta, e, a) = & \max_{\tilde{c}, a'} \left\{ \frac{\tilde{c}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \left[V_{t+1}(s', \beta', e', a' | .) \right] \right\} \\ \text{s.t.} \qquad & \tilde{c} + a' = (1 + r_t) a + d_{i,t} + y_t(s, e) - v(h_t(s, e)) \\ & a' \in [\underline{a}, \infty). \end{aligned}$$

Formal workers maximize their utility taking hours as given, so their labor disutility is internalized at the union level, not by individual households. However, informal hours are chosen and therefore included in the Euler Equation. Given their decision, their consumption-equivalent income is

$$w^I e h_t - v(h_t) = w^I e h_t - \psi \frac{h_{i,t} h_{i,t}^{1/\varphi}}{1 + \frac{1}{\varphi}} = \frac{w^I e h_t}{1 + \varphi}.$$

The labor supply risk is assigned by transition probabilities $\mathbb{P}(s' \mid s)$, where formal households may go to informality with probability p_{fi} and they don't go directly to unemployment, while informal households may go to formality with probability p_{if} and may lose their job with probability p_{iu}. Unemployed households go to formality with probability p_{uf} and to informality with p_{ui}.

$$\Pi_s = \mathbb{P}(s' \mid s) = \begin{bmatrix} 1 - p_{fi} & p_{fi} & 0 \\ p_{if} & 1 - p_{if} - p_{iu} & p_{iu} \\ p_{uf} & p_{ui} & 1 - p_{uf} - p_{ui} \end{bmatrix}.$$

In equilibrium, there is a measure of F_t formal households, I_t informal households, U_t unemployed, and BF_t beneficiaries of the BF program, with all steady-state calculations in the Appendix. We set all probabilities in Π_s to match the actual size of each sector.

In this approach, Π_s is held fixed at its steady-state value, a short-run restriction to the intensive margin, treating the sector recomposition as a medium-run phenomenon.

However, we will change this approach to be endogenously determined in equilibrium, in order to capture this recomposition effects of the conditional transfers.

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs produced by formal firms,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t}$. They hire labor at real wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu W_t$, and the real wage equals $w_t = \frac{1}{\mu}$ at every date in steady-state. Firms' dividends are taxed at the same rate τ_ℓ as labor income, so $d_t = (1 - \tau_\ell)(Y_t - w_t L_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor.

We assume sticky prices following Rotemberg (1982), standard in Neo-Keynesian's literature. Such as Hagedorn, Manovskii, and Mitman (2019), the Phillips curve for price inflation is given by

$$\ln(1 + \pi_t) = \kappa \left[\frac{w_t}{Z_t} - \frac{1}{\mu} \right] + \frac{\ln(1 + \pi_{t+1})}{1 + r_{t+1}} \cdot \frac{Y_{t+1}}{Y_t},$$

Additionally, there is a representative informal firm operating in a perfectly competitive market and producing under constant returns to scale, so $Y_t^I = w_t^I L_t^I$. The steady-state wage gap $\xi = w_t^I / w_t$ is calibrated to match PNAD Contínua data.

Unions

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to Rotemberg (1982)'s adjustment costs. We assume that unions allocate all formal labor hours uniformly across agents, so $h_{i,t} = h_t$ (Auclert, Rognlie, and Straub 2025).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $\bar{C}_t = \int_0^1 c_{it} \, di$, discounted by the average discount factor $\bar{\beta} = \lambda_I \beta_L + (1 - \lambda_I)\beta_H$ (Auclert, Rognlie, Souchier, et al. 2021). This results in a Phillips curve for wage inflation $\pi_t^w = (1 + \pi_t) w_t / w_{t-1}$ given by

$$\ln(1 + \pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} \cdot \tilde{C}_t^{1/\gamma} - \frac{(1 - \tau_\ell) w_t L_t}{\mu_w} \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$

The Phillips Curve calculations for price and wage are found in the Appendix.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r^*_{t-1} + \phi\pi_{t-1}$, where r^* is the neutral real interest rate, and $\phi > 1$ (Taylor 1993).

In the governmental sphere, we have a government that purchase public goods, collects taxes, makes transfers, and incurs debt; therefore, the government's budget constraint is characterized by

$$G_t + T_t \cdot \text{BF}_t + (1 + r_t)B_{t-1} = B_t + \tau_t Y_t,$$

which means that, in steady-state, $G = \tau wL - rB - T \cdot \text{BF}$.

For dynamic simulations, we intend to apply two regimes: one financed by debt, with T fixed and $B_t = (1 + r)B + T \cdot \text{BF} - \tau wL - G$, and another financed by taxes, with B fixed and T variable. For now, we consider fixed T_t and B_t, treated as exogenous, with G_t adjusting to it.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). We define $\Lambda_t(s, \beta, e, a)$ as the household distribution over a given period of time. First, the asset market reaches equilibrium when household savings equal government bonds

$$A_t = \int a_{i,t}\, d\Lambda_t(s, \beta, e, a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$\begin{aligned} L_t &= \int \mathbb{1}\{s_{i,t} = F\} \cdot e_{i,t} \cdot h_{i,t}\, d\Lambda_t(s, \beta, e, a); \\ L^I_t &= \int \mathbb{1}\{s_{i,t} = I\} \cdot e_{i,t} \cdot h_{i,t}\, d\Lambda_t(s, \beta, e, a); \\ \text{and} &\qquad F_t + I_t + U_t = 1; \end{aligned}$$

the BF benefit clears when it is given to the eligible households

$$\text{BF}_t = U_t + \int \mathbb{1}\{s_{i,t} = I, e_{i,t} < e^*\}\, d\Lambda(s, \beta, e, a);$$

and the goods market clear when $Y + Y^I = C + G$ in the steady-state. However, due to the GHH utility, we have $C = \tilde{C} + \int v(h)\, d\Lambda_t(.)$ and

$$\int \psi \frac{h_{i,t}^{1+\frac{1}{\varphi}}}{1 + \frac{1}{\varphi}}\, d\Lambda_t(s, \beta, e, a) = \int \frac{\varphi h_{i,t} w^I_t e_{i,t}}{1 + \varphi}\, d\Lambda_t(s, \beta, e, a) = \frac{\varphi w^I_t L^I_t}{1 + \varphi} = \frac{\varphi}{1 + \varphi} Y^I_t.$$

Thus,

$$Y_t + Y_t^I = \int \tilde{c}_{i,t}\, d\Lambda_t(s, \beta, e, a) + G_t = \tilde{C}_t + \frac{\varphi}{1+\varphi} Y_I + G_t.$$

3. Calibration

Data

To calibrate this model, we use Brazilian data and standard parameters in HANK Literature. Most of the data refers to their average values in 2025.

The labor market moments were obtained in Pesquisa Nacional por Amostra de Domicílio Contínua (PNAD-C), an annual survey reporting formality status, wages, weekly worked hours, etc. This panel is interesting to match mainly the sector sizes and the worked hours, for both formal and informal households. However, since it is a survey, it might have under-reporting of informal firms.

Some general macroeconomic measures were also used, such as the Government Debt (in the Brazilian National Treasure), the interest rates, and the inflation, available in the Índice Brasileiro de Geografia e Estatística (IBGE). The Gini coefficient of wealth was obtained from the 2025 Global Wealth Report, from UBS. Bolsa Família data is available in Cadastro Único (CadÚnico), a platform with several statistics about the benefit.

In the future, we aim to gather information also in the Relação Anual de Informações Sociais (RAIS), a mandatory annual reporting of firms, but without informality information and only up to 2003, and Economia Informal Urbana (ECINF), a survey conducted by IBGE that focus on small business, specially informal ones.

Method and Calibration

The main method is the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, Rognlie, and Straub 2021). The central idea is to summarize the model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones.

The method is built to find General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G}\, d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1} \mathbf{F}_{\mathbf{Z}}\, d\mathbf{Z}.$$

We are interested in calculating $dC = \mathbf{G}_{C,T}\, dT$, the total effect of the change in the conditional transfer on household consumption.

We resolve the heterogeneity by the Endogenous Grid Method (EGM) (Carroll 2006), which solves the value function iteration using the Euler equation, without solving the household maximization problem. Each model period stands for one quarter and the computational time horizon is 100 quarters (or 25 years).

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$,

Table 1: External Calibration

	Description	Value	Source
Households			
γ	EIS	0.5	Standard in Literature
δ_β	Difference: $\beta_H - \beta_L$	0.12	Approximate Gini Coefficient
ω_I	Share of Impatient Agents	0.50	Auclert, Rognlie, and Straub (2025)
Productivity and Asset Grid			
$\underline{a}$	Borrowing constraint	0.0	No Borrowing (McKay and Reis 2016)
$\bar{a}$	Maximum asset	200	Sufficient to cover the top of dist.
N_a	Number of Asset Grid	200	Accuracy near constraint
ρ_e	Persistence of log-productivity	0.966	Kaplan, Moll, and Violante (2018)
σ_e	Std. dev. of productivity	0.85	Bartal and Becard (2024)
N_e	Number of Nodes	15	Standard Discretization
Firms			
Y	Aggregate Output	1.0	Normalization
ξ	Wage gap $= w^I/w$	0.64	PNAD-C Data
μ	Price Markup	1.11	Basu and Fernald (1997)
μ_w	Price Markup	1.11	Equal to μ
κ	Price-NKPC Slope	0.025	Hazell et al. (2022)
κ_w	Wage-NKPC slope	0.025	Equal to κ
Government and Monetary Policy			
r^*	Neutral Real Interest Rate	1.0%	4% annual average
ϕ	Taylor Rule Coefficient	1.5	Standard (Taylor 1993)
τ_ℓ	Labor Income Tax Rate	0.27	Average Tax Payroll in Brazil
T/w	Bolsa Família Transfer	0.14	CadÚnico, PNAD-C

Table 2: Endogenous Calibration

	Description	Value	Target	Model	Data	Source
Households						
ψ	Disutility of Labor	0.30	h_F	1.0	1.0	Normalization
φ	Frisch Elasticity	0.15	$\mathbb{E}(h_I)$	0.89	0.87	PNAD-C Data
β_H	Patient's Discount Factor	0.966	B/Y	3.2	3.2	Tesouro Nacional
Labor market						
p_{fi}, p_{uf}	Formal Transition	$0.11, 0.05$	F	50.5%	50.0%	PNAD-C Data
p_{if}, p_{iu}	Informal Transition	$0.12, 0.02$	I	40.0%	44.2%	PNAD-C Data
p_{ui}	Unemployed Transition	0.11	U	5.5%	5.8%	PNAD-C Data
$\bar{y}$	Bolsa Família Threshold	1.0	BF	0.321	0.358	CadÚnico, PNAD-C

with $\varepsilon^e \sim N(0, \sigma_e^2)$. This is approximated by a finite Markov chain with N_e grid points (Rouwenhorst 1995), which is more accurate than Taucher for highly persistent processes. We also assume a log-spaced grid for assets, in order to generate more robustness near the credit constraint.

The discount factor transition (β_L, β_H) is modeled by a redrawing probability of $q = 0.10$, corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025). This aims to capture similar results in relation to a life-cycle overlapping generations with permanent ability shock model.

We solve the steady-state equilibrium computationally by Brent's method, internally calibrating parameters in order to clear the asset market, the labor market, the government budget constraint, and the wage Phillips curve. The external calibration is shown in table 1 and the endogenous calibration is in 2. The goods market was left untargeted to verify the Walras' Law.

In the household preferences, elasticity of intertemporal substitution is standard in literature $\gamma = 0.5$[2]. The disutility of labor $\psi = 0.30$ is internally calibrated to normalize formal hours $h_F = 1.0$. At last, $\varphi = 0.15$ is chosen to approximate informal worked hours to PNAD-C Data, in average (model = 0.891, data = 0.867) and in standard error (model = 0.35, data = 0.34)[3].

The labor market transition probabilities are defined to match the sector sizes in PNAD-C, $F = 50.5\%$ (data = 50.0%), $I = 44.0\%$ (data = 44.2%), and $U = 5.0\%$ (data = 5.8%). Their values are $p_{fi} = 0.11$, $p_{if} = 0.12$, $p_{ui} = 0.11$, $p_{uf} = 0.05$, and $p_{iu} = 0.02$. In the future, I am interested in estimating the sectoral transition risk in the PNAD-C, if feasible.

The output is normalized $Y = 1.0$ and we defined $\mu = 1.11$, which is standard in the HANK literature (Kaplan, Moll, and Violante 2018; Auclert, Rognlie, and Straub 2025)[4], and $\kappa = \kappa_w = 0.025$, consistent with Hazell et al. (2022) estimates.

Taylor Rule is pretty standard, with $\phi = 1.5$ and $r^* = 1\%$, consistent with a 4% annual real interest rate for Brazilian economy (Santos and Muinhos 2024). The Government debt is set to 80% of yearly GDP, consistent with Brazilian National Treasure data. Wage tax is an average of all contributions levied on the payroll, $\tau_\ell = 0.27$.

The Bolsa Família statistics were obtained from the Cadastro Único[5], where we calibrated the T/w ratio in steady-state to be equal to the data from the Brazilian economy, where $T =$ R\$ 711 and $w =$ R\$ 4,183. We also calibrated the Bolsa Família threshold $\bar{y}$ to match the actual size of the program [6].

[2] EIS $\gamma = 0.5$ is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).

[3] We used the average of half the sample because, in the future, high-productivity informal workers will choose for formalization.

[4] Markup $\mu = 1.11$ is also consistent with the estimates of Basu and Fernald (1997).

[5] Available in VIS DATA 3 and Cadastro Único.

[6] We used the average household size in PNAD-C to calculate a percentage estimate of the program.

4. Results

We solve the model at the calibrated steady-state and compute both partial-equilibrium (PE) and general-equilibrium (GE) Jacobians with respect to the BF Transfer T, treating it as the fiscal instrument.

Steady-State distribution. The wealth distribution is highly concentrated near the borrowing constraint, with 14.3% of households holding zero liquid assets (hand-to-mouth). The model Gini Coefficient for wealth is 0.52, below the SCF 2019 US benchmark (0.77)[7] and the Brazilian estimation from UBS (0.82), consistent with a one-asset structure that compresses the upper tail. In the future, we will add extra features to increase wealth inequality in the model.

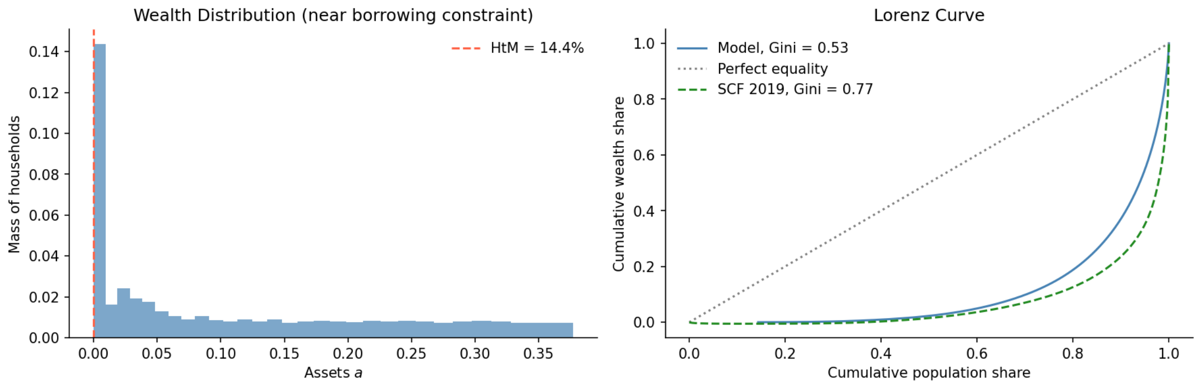

Figure 1: Wealth Distribution

Consumption policy function. Policy functions are nearly linear in assets across employment states. Formal, informal, and unemployed households have roughly parallel schedules, with the informal and unemployed shifted down due to lower income. In this setup, formal and informal have almost the same consumption function, due to the BF benefit. At high asset levels, the slope approaches r, indicating that wealthy households are near permanent-income consumers and save most of any windfall.

[7]The US SCF 2019 was added solely to allow for some comparison.

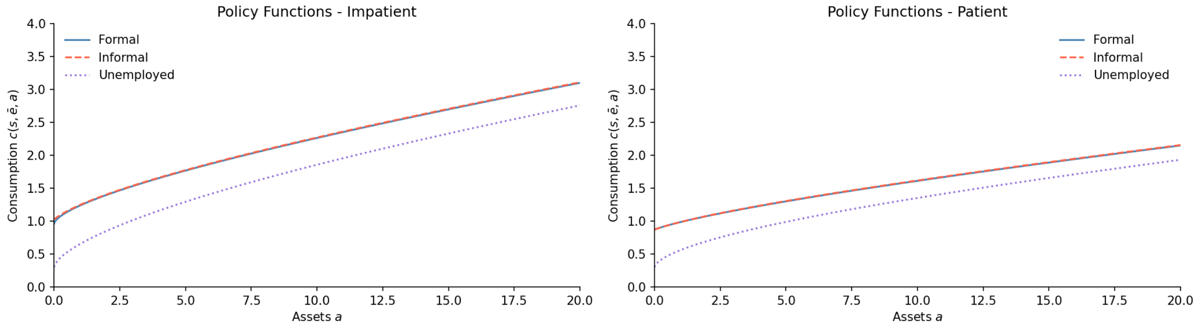

Figure 2: Steady-State Consumption Policy Function

Intertemporal MPC profile. The PE iMPC of aggregate consumption with respect to T is large on impact (8% per unit transfer) and decays monotonically, nearly reaching zero after three years. The rapid decay reflects the high share of constrained BF recipients who spend immediately.

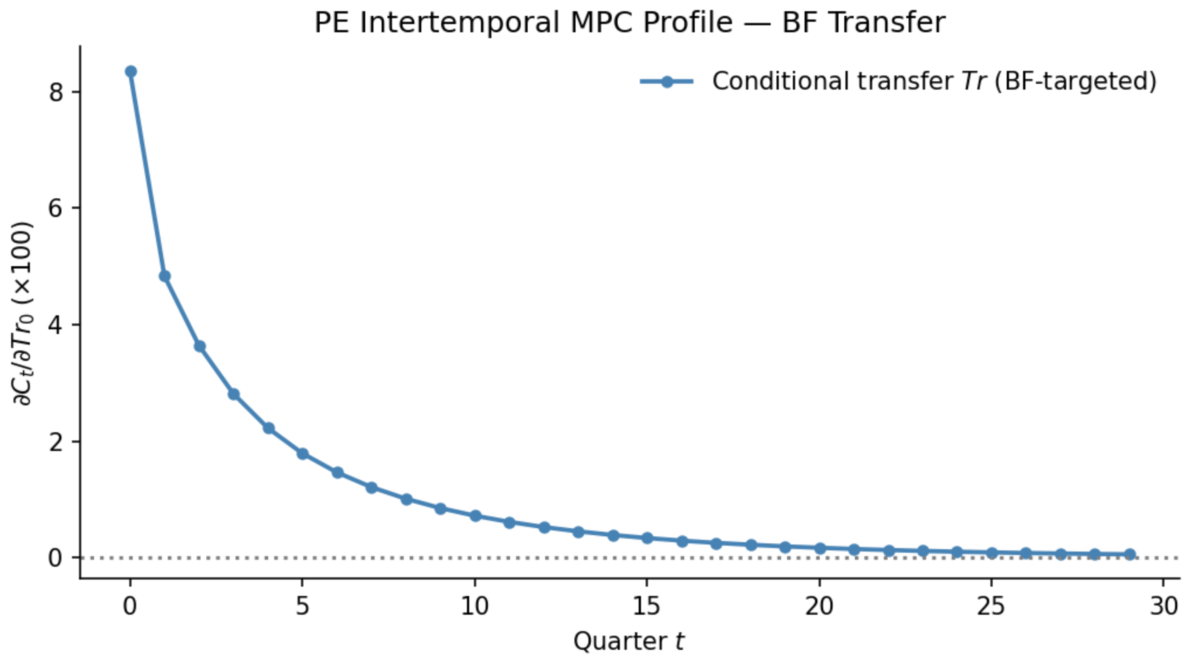

Figure 3: Partial Equilibrium iMPC Profiles

General equilibrium impulse responses. Under a persistent 1% rise in T (AR(1), $\rho = 0.4$), output and consumption GE responses track the PE responses closely. GE amplification through the wage and interest rate channels is negligible at the current calibration. The BF recipient count does not respond to the shock, since labor market transitions are exogenous, a restriction we will change.

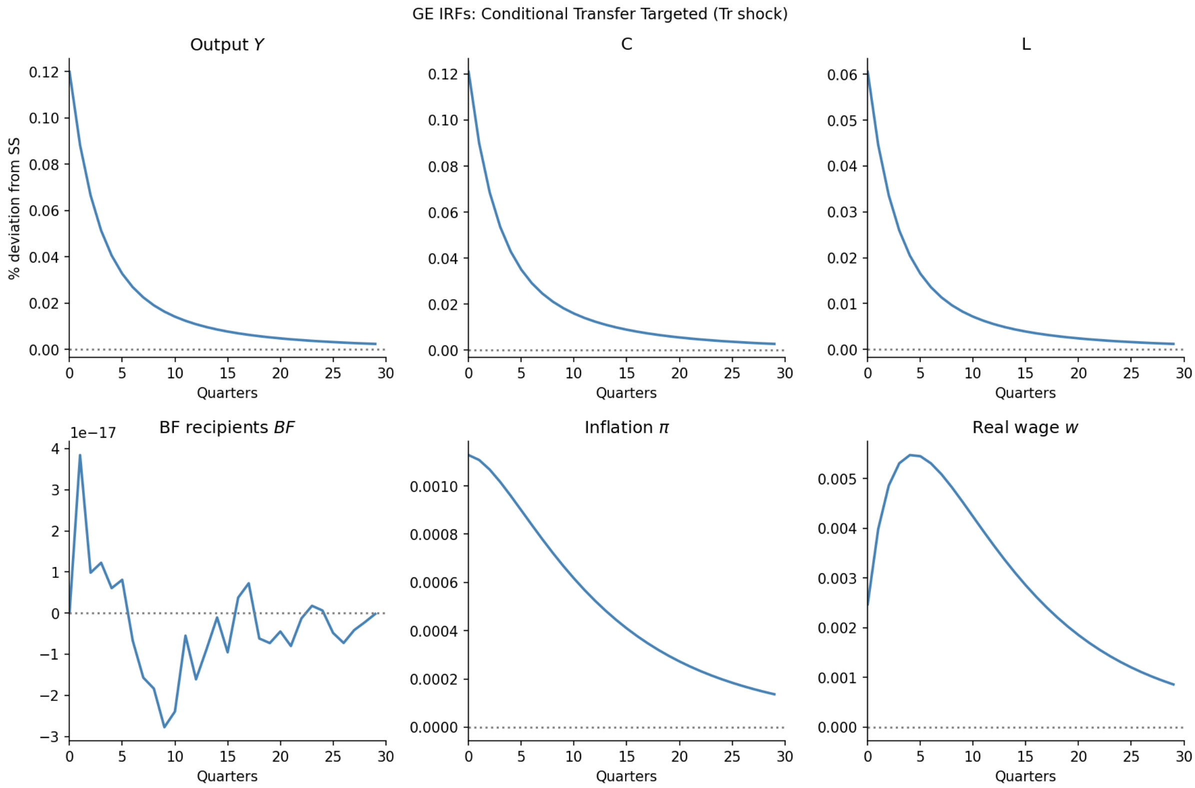

Figure 4: IRFs: BF-Targeted (shock in T)

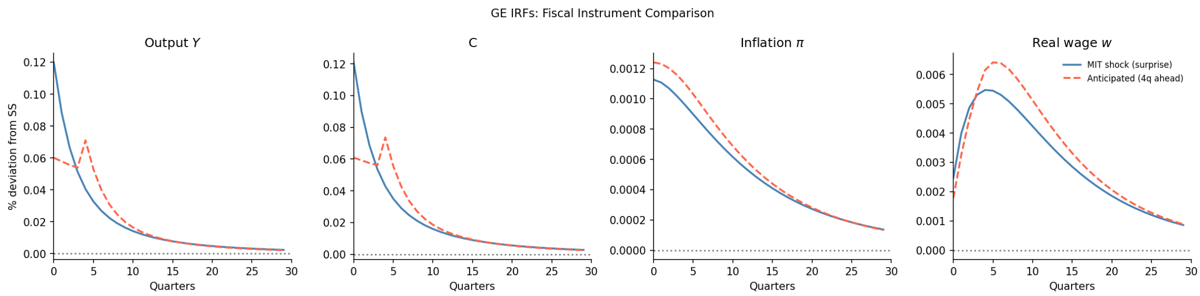

Figure 5: IRFs: Comparison (shock in T, MIT vs Antecipated)

Robustness Tests

Lastly, in order to verify whether this results are representatives and robust, we intent to do some robustness tests changing some calibration parameters, such as γ, δ_β, ω_I, ξ, μ, and κ_w. This parameters might have minor implications in the results, but it is expected that the model continues to have the same results.

Further, since there are several combinations of the sector transition probabilities that implies in the same sector sizes, we aim to changing each of this probability and capture the result. This change will affect the dynamics and may alter the transfer effects, due to a faster or slower transition to the informal sector. If the transition data is feasible in PNAD-C, it will not be necessary.

Also, as we commented early, we intent to perform other dividend specifications instead of $d_{i,t} = d_t \cdot e_{i,t}$. We also aim to change the GHH utility to a standard separable utility in the form of $u(c) - v(h)$, which must have similar results.

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Appendix

Sectors Size in Steady-State

The steady-state level of each sector satisfies $\pi^s \Pi_s = \pi^s$, where π^s is the steady-state matrix $[F, I, U]$. Resolving this system, plus the normalization $F + I + U = 1$,

$$\begin{bmatrix} (1-p_{fi})F + p_{if}I + p_{uf}U \\ p_{fi}F + (1-p_{if}-p_{iu})I + p_{ui}U \\ p_{iu}I + (1-p_{uf}-p_{ui})U \end{bmatrix} = \begin{bmatrix} F \\ I \\ U \end{bmatrix},$$

which implies that

$$U = \underbrace{\frac{p_{iu}}{p_{uf}+p_{ui}}}_{\gamma} I; \qquad F = \underbrace{\frac{p_{if}(p_{uf}+p_{ui}) + p_{iu}p_{uf}}{p_{fi}(p_{uf}+p_{ui})}}_{\alpha} I.$$

Thus, with $F + I + U = 1$,

$$F = \frac{\alpha}{1+\alpha+\gamma}; \qquad I = \frac{1}{1+\alpha+\gamma}; \qquad U = \frac{\gamma}{1+\alpha+\gamma}.$$

This means that we can fix three transition rates and then resolve for the other two analytically, given the three sector sizes.

Price Phillips Curve

Let Y_t be the aggregate output of the economy, a CES of the intermediary inputs,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} \, dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

This firm maximizes its output subject to $\int_0^1 p_{i,j} y_{i,j} \, dj$. The FOC implies that the demand for the intermediate good is given by

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t,$$

where P_t is the equilibrium price of the final good,

$$P_t = \left(\int_0^1 p_{j,t}^{1-\varepsilon} \, dj \right)^{\frac{1}{1-\varepsilon}}.$$

We assume a Rotemberg (1982) quadratic price adjustment cost given by

$$\frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t,$$

where $\theta_p = (\varepsilon - 1)/\kappa$ is the adjustment cost multiplier and $\bar{\Pi}$ the steady-state inflation.

Let $\Xi(y)$ be the intermediary firm's cost. Given last period's price $p_{j,t-1}$, the firm chooses this period's price $p_{j,t}$ to maximize the present discounted values of future profits,

$$V_t^{IF}(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} y_{j,t} - \Xi(y_{j,t}) - \frac{\theta_p}{2}\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}^{IF}(p_{j,t}).$$

We can replace the demand function into this problem, so

$$V_t(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t}\left(\frac{p_{j,t}}{P_t}\right)^{-\varepsilon} Y_t - \Xi(y_{j,t}) - \frac{\theta_p}{2}\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}(p_{j,t}).$$

The FOC with respect to $p_{j,t}$ is

$$(1-\varepsilon)\left(\frac{p_{j,t}}{P_t}\right)^{-\varepsilon}\frac{Y_t}{P_t} + \varepsilon mc_{j,t} - \theta_p\left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi}\right)\frac{Y_t}{p_{j,t}} + \frac{1}{1+r_{t+1}} V'_{t+1}(p_{j,t}) = 0,$$

where $mc_t = \partial\Xi/\partial y_{j,t}$, and the envelope condition gives us

$$V'_{t+1} = \theta_p\left(\frac{p_{j,t+1}}{p_{j,t}} - \bar{\Pi}\right)\frac{p_{j,t+1}}{p_{j,t}}\frac{Y_{t+1}}{p_{j,t}}.$$

Thus, taking $p_{j,t} = P_t$ in equilibrium,

$$\theta_p(\pi_t - \bar{\Pi})\pi_t = \varepsilon\left(mc_t - \frac{\varepsilon-1}{\varepsilon}\right) + \frac{1}{1+r_{t+1}}\theta_p(\pi_{t+1} - \bar{\Pi})\pi_{t+1}\frac{Y_{t+1}}{Y_t},$$

where we define the firm's markup as $\mu = \frac{\varepsilon}{\varepsilon-1}$. Linearizing around the steady-state $\pi_t = \bar{\Pi}$,

$$\ln(1+\pi_t) = \frac{\varepsilon}{\theta_p}\left(mc_t - \frac{1}{\mu}\right) + \frac{1}{1+r_{t+1}}\ln(1+\pi_{t+1})\frac{Y_{t+1}}{Y_t},$$

where $\kappa = \varepsilon/\theta_p$ and $mc_t = w_t/Z_t$. This environment is equivalent to Calvo (1983) in first order around zero inflation.

Wage Phillips Curve

We will use the same setting as Hagedorn, Manovskii, and Mitman (2019), but considering just the formal sector. Assume there is a union that aggregate effective labor input into

$$h_t = \left(\int_0^1 e_{i,t}(h_{i,t})^{\frac{\varepsilon_w - 1}{\varepsilon_w}}\, di\right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}},$$

sells households labor services $e_{i,t}h_{i,t}$ at nominal wage $W_{i,t}$ to the labor recruiter, which minimizes costs $\int_0^1 W_{i,t}e_{i,t}h_{i,t}\, di$, implying a demand of

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t}\right)^{-\varepsilon_w} h_t.$$

The unions sets a nominal wage $\hat{W}_t$ for an effect unit of labor, $W_{i,t} = \hat{W}_t$ to maximize profits subject to Rotemberg (1982)'s wage adjustment costs

$$e_{i,t}\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}}-\bar{\Pi}^w\right)^2 h_t.$$

We assume that the union's wage setting problem is

$$V_t^w = \max_{\hat{W}_t}\int\left[\frac{(1-\tau_\ell)\hat{W}_t}{P_t}e_{i,t}h_{i,t}-\frac{v(h_{i,t})}{u'(\tilde{C}_t)}-e_{i,t}h_t\frac{\theta_w}{2}\left(\frac{\hat{W}_t}{\hat{W}_{t-1}}-\bar{\Pi}^w\right)^2\right]d\Lambda+\bar{\beta}V_{t+1}^w.$$

Using the same algebra as the price Phillips curve, assuming that the union allocates the same amount of hours in each worker $h_{i,t}=h_t$, and using $\int_F e_{i,t}d\Lambda = L_t$, it is possible to find that

$$\theta_w(\pi_t^w-\bar{\Pi}^w)\pi_t^w=\varepsilon_w\left[\frac{v'(h_t)}{u'(\tilde{C}_t)}-\frac{\varepsilon_w-1}{\varepsilon_w}(1-\tau_\ell)w_tL_t\right]+\bar{\beta}\theta_w(\pi_{t+1}^w-\bar{\Pi}^w)\pi_{t+1}^w\frac{h_{t+1}}{h_t},$$

which we can linearize around the steady state $\pi_t^w=\bar{\Pi}^w$ and take $h_{t+1}=h_t$, since there is no population nor technological growth. Thus,

$$\ln(1+\pi_t^w)=\frac{\varepsilon_w}{\theta_w}\left[\frac{v'(h_t)}{u'(\tilde{C}_t)}-(1-\tau_\ell)w_tL_t\frac{\varepsilon_w-1}{\varepsilon_w}\right]+\bar{\beta}\ln(1+\pi_{t+1}^w).$$

Therefore, defining $\kappa_w=\varepsilon_w/\theta_w$ and $\mu_w=\varepsilon_w/(\varepsilon_w-1)$, the wage Phillips curve is

$$\ln(1+\pi_t^w)=\kappa_w\left[\psi\cdot h_t^{1/\varphi}\cdot\tilde{C}_t^{1/\gamma}-\frac{(1-\tau_\ell)w_tL_t}{\mu_w}\right]+\bar{\beta}\ln(1+\pi_{t+1}^w).$$